

Project Proposal: EEG Neurotech Mobile App (Adapting SMART Hydroponic to g.tec Unicorn)

CSE 431: Mobile Programming, Fall 2025
Presented to Dr. Haytham Azmi

October 2025

1 Introduction

This proposal adapts the “SMART Hydroponic” mobile application specification to our graduation project domain: a real-time EEG neurotechnology app using the g.tec Unicorn Hybrid Black headset. We map the hydroponic system’s sensors, actuators, cloud connectivity, analytics, alerts, and testing requirements to EEG acquisition, analytics, and neurofeedback control. Our goal is to meet all functional and non-functional requirements to achieve full marks, while producing a great front-end for our graduation project.

Please note that the core features of the GP are the data-acquisition, pre-processing, and data-analysis (DL) pipeline. We thought it may be a good idea to “merge” the mobile-programming project also to add a good front-end for the GP.

2 Objective and concept

Objective: Build a Flutter app that:

- **Monitors** real-time EEG signals and derived features (e.g., band power, cognitive workload level, detected emotion).
- **Provides** neurofeedback control for mental-state regulation (visual, auditory). (e.g progress bars, music loudness)
- **Synchronizes** data with Firebase for dashboards and remote review, while logging locally in SQLite.
- **Analyzes** history & trends. Provides over-time insights and threshold-triggered notifications.

3 Requirement mapping (Hydroponic $\rightarrow$ EEG)

3.1 Functional screen design

The functional screen will be very similar to the one outlined in the Hydroponic project document, and will have the same functionalities (e.g registration & authentication, real-time sensor monitoring, analytics & history, etc) with minor domain-specific differences.

3.2 Hardware and mobile technology

- **Sensor hub (MCU equivalent):** g.tec Unicorn Hybrid Black streams EEG wirelessly to a host (PC bridge).
- **Sensors:** 8 EEG channels (primary sensors) plus derived “virtual sensors”:
- **Actuators:**
 - **Start/Stop Recording Session**
 - **Neurofeedback Control:** Color/brightness modulation, progress bar animation driven by thresholds, etc.
- **Cloud link (Firebase):** Publish analyzed data for storage, receive control commands.
- **Architecture:**
 - **PC bridge :** Flutter app issues commands via Firebase/Socket; a lightweight PC service (Unicorn SDK) controls acquisition and mirrors live features to Firebase; mobile displays/sends feedback.

3.3 Database integration

We will implement SQLite for local logging and state management with Provider/Bloc.

3.4 Quality enhancement features

- **Text-to-speech (Maybe Implemented):** Impedance guidance, coaching prompts, session status changes.
- **Speech recognition (Maybe Implemented):** Hands-free commands: “start session”, “pause feedback”, etc.
- **Notifications (Will be Implemented):** High workload, low SNR, connection loss, schedule start/end, session summary.
- **MVVM architecture (Will be Implemented)**
- **Data validation (Will be Implemented):** Numeric thresholds, impedance acceptable before “Start”.

3.5 Testing and auto-test scripts

We will provide full testing and automation as outlined in the requirements document.

4 Conclusion

This adaptation satisfies the SMART Hydroponic specification by mapping environmental sensors/actuators to EEG features and neurofeedback, preserving cloud synchronization, local logging, analytics, alerts, and testing. It simultaneously advances our graduation project by delivering a production-quality mobile UI, session orchestration, and data management, aligned with the marking criteria for full credit.

Final Note: If some of the project's core functionalities (e.g data-analysis unit, deep learning model) are not ready by the time of the app's delivery time, we will use simulated data and/or drivers and stubs to meet the full requirements of the course CSE431.